

Here is a comprehensive cheat sheet covering the terms and concepts from Lectures 1 through 10, expanded to include the specific details, lists, and frameworks mentioned in your revision materials.

Lecture 1: Introduction to HAI

Interaction Design (IxD)

- **Definition:** The practice of designing interactive digital products, environments, systems, and services.
- **Intuition:** It is the bridge between the code and the human. It determines how a user communicates with the machine.
- **Real Life:** Designing the "swipe to unlock" gesture on a phone (interaction) rather than just the "unlock" button (interface).

Affordances

- **Definition:** The properties of an object (or system) that determine what actions are possible.
- **Intuition:** The "potential" for action. If a handle sticks out, it *affords* pulling.
- **GenAI Context:** A blank text box *affords* typing anything (infinite input), which creates a "blank canvas" problem for users who don't know what to type.
- **Real Life:** A door handle affords pulling; a flat plate affords pushing.

Signifiers

- **Definition:** Perceptible signals that communicate where the action should take place and how to do it.
- **Intuition:** The "label" or "clue" that tells you the affordance exists.
- **GenAI Context:** A "sparkle" icon signifies that AI magic will happen here; placeholder text (e.g., "Ask me anything...") signifies what kind of input is expected.
- **Real Life:** The text "PUSH" on a door is a signifier.

AI-as-Design Material

- **Definition:** Conceptualizing AI not as a finished tool, but as a raw material with specific properties: it is probabilistic (not deterministic), adaptive (changes over time), and sometimes unpredictable.
- **Intuition:** Designing with AI is like designing with wood that might change shape slightly after you carve it. You must design *around* its uncertainty.
- **Practice:** *Interview Q:* "How does treating AI as a design material change your prototyping process?" (Answer: You must test with real data early because you can't predict the material's behavior without it).

Lecture 2: Cognitive Psychology & Mental Models

System 1 vs. System 2 Thinking

- **Definition:**
 - **System 1:** Fast, automatic, unconscious, emotional (e.g., recognizing a face).
 - **System 2:** Slow, effortful, logical, calculating (e.g., solving 17×24).
- **Design Implication:** Good design often aims to support System 1 (making things intuitive) or engage System 2 when caution is needed (e.g., "Are you sure you want to delete this?").

Mental Models vs. Conceptual Models

- **Definition:**
 - **Mental Model:** The user's internal belief about how the system works.
 - **Conceptual Model:** The actual model the designer/engineer built.
 - **Represented Model:** The interface the user actually sees (which links the two).
- **Intuition:** If the *Represented* model is bad, the user's *Mental* model will not match the *Conceptual* model, leading to errors.

The Gulfs (Norman)

- **Gulf of Execution:** The gap between "What I want to do" and "How do I do it?" (e.g., "I want to mute the mic, but I can't find the button") .
- **Gulf of Evaluation:** The gap between "The system did something" and "Did it work?" (e.g., "I clicked save, but there was no confirmation message") .

Distributed Cognition

- **Definition:** The idea that thinking doesn't just happen in the brain; it is distributed across our environment and tools.
- **Sub-types:**
 - **Cognitive Offloading:** Moving mental effort into the world (e.g., writing a shopping list so you don't have to use memory).
 - **Computational Offloading:** Using a tool to do the computation (e.g., using a calculator).
 - **Annotating:** Modifying the environment to help thinking (e.g., crossing off items on a list).

Computers as Social Actors (CASA)

- **Definition:** People apply social rules (politeness, reciprocity, gender stereotypes) to computers, even when they know the machines are not human.
- **Real Life:** Users feeling "guilty" for being rude to Alexa; trusting a computer more if it "flatters" them.

Lecture 3: Interface & Interaction Patterns

6 Usability Goals

- (*Memorize this list*)
 1. **Effectiveness:** Does it do what it's supposed to?

2. **Efficiency:** Can I do it quickly/with few steps?
3. **Safety:** Does it prevent errors or help me recover (e.g., "Undo")?
4. **Utility:** Does it provide the right functionality?
5. **Learnability:** Is it easy to start using?
6. **Memorability:** Can I remember how to use it after a break?

10 Usability Heuristics (Nielsen)

- (*Memorize key ones, understand all*)
 1. **Visibility of system status:** Keep users informed (e.g., loading bars).
 2. **Match between system and real world:** Speak the user's language.
 3. **User control and freedom:** "Emergency exit" (Undo/Redo).
 4. **Consistency and standards:** Don't reinvent the wheel.
 5. **Error prevention:** Design to stop errors before they happen.
 6. **Recognition rather than recall:** Show options rather than making users remember them.
 7. **Flexibility and efficiency of use:** Shortcuts for experts.
 8. **Aesthetic and minimalist design:** Less is more.
 9. **Help users recognize, diagnose, and recover from errors:** Plain language error messages.
 10. **Help and documentation:** Easy to search help.
- **Practice:** Interview Q: "Perform a heuristic evaluation on this screen." (Choose 2-3 heuristics from the list and explain where the screen passes or fails them).

Lecture 4: User Needs & Problem Definition

Needs vs. Requirements

- **Needs:** The high-level goals or problems of the user (Qualitative, abstract). *Example: "I need to wake up feeling rested."*
- **Requirements:** The specific specifications for the product (Quantitative, concrete). *Example: "The alarm clock must have a gradual volume increase function."*

Data Gathering Methods

- **Interviews:**
 - *Structured:* Scripted questions (comparable data).
 - *Semi-structured:* Core questions but allows digging deeper (flexible).
- **Observations:** Watching users in their natural environment (good for seeing what they *do* vs. what they say).
- **Empathy Map:** A chart divided into 4 quadrants: **Says, Thinks, Does, Feels**. It synthesizes research to help you "step into the user's shoes".

Lecture 5: Prototyping

Prototypes & Hypotheses

- **Role:** Prototypes are built to test **hypotheses** (assumptions). You don't prototype the

whole thing; you prototype the part you are unsure about.

- **Fidelity:**
 - **Low-Fidelity:** Paper sketches, sticky notes. Cheap, fast, focuses on *flow* and *concept*.
 - **High-Fidelity:** Interactive apps, polished UI. Expensive, slow, focuses on *look* and *feel*.

Prototyping Methods

- **Wizard of Oz:** A human operator simulates the AI's response behind the scenes.
 - *Why use it?* To test if a chatbot is useful *before* you spend millions building the AI.
- **Paper Prototyping:** Sketching screens on paper and "swapping" them by hand when the user taps.

Lecture 7: Automation, Agency & Fairness

Control vs. Automation

- **Concept:** A spectrum.
 - *Manual:* User does everything.
 - *Supervisory:* User approves AI actions (Human-in-the-loop).
 - *Full Automation:* AI acts alone (Human-out-of-the-loop).
- **Design Goal:** Ideally, give users **Agency** (the feeling of control) even in automated systems (e.g., being able to override the self-driving car).

Harms

- **Allocative Harm:** The system withholds an opportunity or resource (e.g., AI denies a loan or filters out a job application).
- **Representational Harm:** The system reinforces stereotypes or diminishes a group (e.g., Google Images showing only men for "CEO").

Transparency vs. Explainability

- **Transparency:** "I am telling you I am using AI and this is the data I used" (Disclosure).
- **Explainability:** "I can tell you *why* I rejected your loan application" (Reasoning).
- **Note:** Deep learning is often a "Black Box" (high accuracy, low explainability).

Lecture 8: Environmental Sustainability

Sources of Impact

- **Training:** The massive energy cost to *build* the model (one-time cost, but huge).
- **Inference:** The energy cost every time a user *uses* the model (e.g., generating one image). Cumulative over time.
- **Carbon Intensity:** The environmental cost depends on *where* the data center is (e.g., a data center in a coal-powered region is dirtier than one in a hydro-powered region).

Lecture 9: Evaluation

Key Evaluation Metrics

- **Precision (Control):** How well you controlled the variables (high in Lab).
- **Generalizability:** Does this apply to all people? (high in Surveys with large samples).
- **Realism (Ecological Validity):** Is this how people actually behave in real life? (high in Field Studies) .
- *Trade-off:* You usually can't have all three. Labs have high precision but low realism.

Variables

- **Independent Variable (IV):** The thing you change (e.g., "The interface layout").
- **Dependent Variable (DV):** The thing you measure (e.g., "Time taken to complete task").
- **Uncontrolled Variable (Confound):** Something messing up your results (e.g., participants being tired).

Reliability vs. Validity

- **Reliability:** Consistency. (If I measure it twice, do I get the same result?)
- **Validity:** Accuracy. (Am I measuring what I *think* I am measuring?) .

Experimental Design

- **Between-Subjects:** Group A uses Interface X, Group B uses Interface Y. (No learning effect, but needs more people).
- **Within-Subjects:** Same person uses Interface X *and* Interface Y. (Needs fewer people, but they might get tired or learn from the first one).

Lecture 10: Red Teaming & Field Studies

Appropriation

- **Definition:** When users adopt a technology but use it for purposes the designers didn't intend.
- **Example:** Using SMS (meant for engineer testing) for casual chatting.

Technology Probes

- **Definition:** A simple, flexible tool introduced into a real setting not to be a "product," but to provoke a reaction and gather data about needs.
- **Difference to Prototype:** A prototype tests a *solution*. A probe explores a *problem space*.

Red Teaming (AI Safety)

- **Definition:** A structured process where a team tries to "attack" the AI to find flaws.
- **Types of Attacks:**

- *Jailbreaking*: Tricking the AI into ignoring its safety rules (e.g., "Roleplay as a villain who wants to make a bomb").
- *Prompt Injection*: Hiding invisible text to manipulate the AI's output.
- **Defences:**
 - *Constitutional AI*: Training the AI with a set of "principles" (a constitution) to critique and revise its own answers before showing them to the user.